BRAINWARE UNIVERSITY

ML PROJECT-BASED LEARNING REPORT (WEEK 1)

1. Basic Information

Title of the Project:	House Price Prediction System
Name(s) of Student(s):	Subhasis Dhara (BWU/BTS/23/252) Debangshu Biswas (BWU/BTS/23/248) Soumadip Kar (BWU/BTS/23/241)
Programme & Semester:	B.Tech CSE & 6th Semester
Course Name & Code:	Machine Learning Lab & PCC-CSG691A
Name of the Guide/Supervisor:	MR. PRANAB GHARAI & MR. PULAKESH ROY

2. Introduction and Background

House price prediction is a significant application of machine learning that helps estimate property values based on various structural and geographical features. This project aims to build an end-to-end ML-powered full-stack web application using React.js and FastAPI that takes house attributes as input and returns an accurate price prediction.

Week 1 focuses on establishing the project foundation: setting up the development environment, collecting and loading the Housing.csv dataset, and initialising the FastAPI backend. These activities form the base upon which all subsequent data processing, modelling, and deployment work will be built in the coming weeks.

3. Objectives of the Project (Week 1)

- Set up the development environment including Python, Node.js, and required libraries
- Collect and load the Housing.csv dataset from Kaggle
- Initialise the project structure with version control using Git
- Create and configure the FastAPI backend with basic routes and server setup
- Perform an initial data inspection to understand the dataset structure and features
- Define the overall machine learning pipeline and project architecture

4. Problem Statement / Driving Question

How can we accurately predict house prices using historical housing data and machine learning algorithms, and how can this be delivered as a reliable, user-friendly full-stack web application?

5. Methodology (Week 1 Activities)

The following tasks were completed during Week 1:

1. Project Setup:

- Installed and configured the development environment: Python 3.x, Node.js, pip, and npm
- Created the project directory structure separating frontend (React.js) and backend (FastAPI) components
- Initialised a Git repository for version control and collaborative development
- Installed required Python libraries: pandas, numpy, scikit-learn, xgboost, joblib, fastapi, uvicorn

2. Dataset Collection:

- Sourced the Housing.csv dataset from Kaggle containing house features and price labels
- Loaded and inspected the dataset using Pandas to review columns, data types, and initial statistics
- Identified key features: area, bedrooms, bathrooms, stories, parking, furnishing status, and price
- Documented feature descriptions and potential engineering opportunities for Week 2

3. FastAPI Backend Initialisation:

- Created the main FastAPI application file (main.py) with a health-check route
- Configured CORS middleware to allow communication with the React.js frontend
- Set up a /predict endpoint skeleton for future model integration
- Verified the server runs correctly using Uvicorn on localhost

6. Expected Outcomes / Deliverables (Week 1)

- Configured and operational development environment
- Loaded Housing.csv dataset with initial inspection and feature documentation
- Initialised Git repository with project structure
- A running FastAPI backend with basic routes and CORS configuration
- Defined project architecture plan for Weeks 2–4
- Week 1 project report

7. Project Timeline

Week	Activity & Status
Week 1 (Current)	**Project Setup, Dataset Collection & Data Preprocessing — Completed**
Week 2 (Upcoming)	Exploratory Data Analysis, Feature Engineering & Model Building
Week 3 (Upcoming)	Model Training, Model Optimization & Evaluation
Week 4 (Upcoming)	Testing, UI Improvement & Deployment

8. Tools & Technologies Used

Tool / Technology	Purpose
Python 3.x	Primary programming language for ML and backend development
Pandas & NumPy	Dataset loading, inspection, and initial preprocessing
XGBoost	Regression model planned for house price prediction
Scikit-learn	Model evaluation metrics (R^2, MAE, RMSE) — planned
Joblib	Model serialisation for deployment — planned
FastAPI (Python)	Backend REST API framework for serving predictions
React.js	Frontend interactive interface for house price input and display
Git	Version control for collaborative development
Kaggle	Source for the Housing.csv dataset
Google Colab	Interactive environment for initial dataset exploration

9. References

- Scikit-learn Official Documentation: https://scikit-learn.org
- XGBoost Official Documentation: https://xgboost.readthedocs.io
- FastAPI Official Documentation: https://fastapi.tiangolo.com
- React.js Official Documentation: https://react.dev
- Kaggle Housing Datasets: https://www.kaggle.com/datasets
- Brainware University — Machine Learning Course Guidelines

10. Signatures

Signature of Student(s):

Date of Submission:

Signature of Guide/Supervisor: